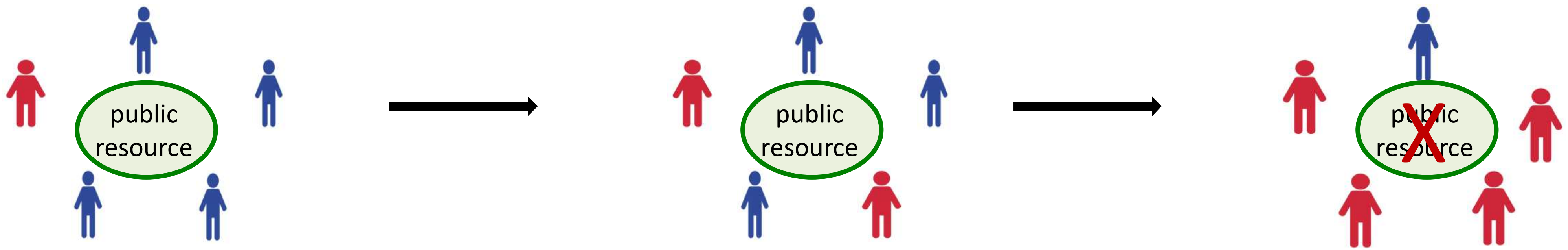
Public Goods Game and the tragedy of the commons



Tragedy of the commons

n: Group size

Payoff: $\pi_{iC} = rkc/n - c$

Payoff: $\pi_{iD} = rkc/n$

k: #Contributors; r: Synergy factor; c: Cost of contributing

Compares her payoff with *another randomly picked* member

Benefits from the public resource without paying the cost for sustaining it

Public Goods Game and the tragedy of the commons

Payoff: $\pi_C = rkc/n - c$ Payoff: $\pi_D = rkc/n$

$$\boxed{\pi_C(k) = \pi_D(k) - c}$$

$$\pi_C(k=n) = c(r-1) \qquad \pi_D(k=0) = 0$$

$$\pi_C(k=n) > \pi_D(k=0) ==> \boxed{r > 1}$$

$$\Downarrow$$

Social dilemma exists for r>1

It is better for ALL to play **C** than for ALL to play **D** even though

$$\pi_C(k) < \pi_D(k) \quad \text{for} \quad 1 \le k \le n-1$$

Linear PGG

- Given a state with ***k*** contributors, is **C**ooperating a more rational choice than **D**efecting i.e. would it be favourable for you to play **C** instead of **D** assuming all other players keep their actions unchanged ?

Yes, if $$\boxed{\pi_C(k+1) - \pi_D(k) > 0}$$

Equivalent to asking if $$\boxed{\pi_C(k+1) - \pi_C(k) > c}$$ since $\pi_D(k) = \pi_C(k) + c$

The increase in payoff to all **C**'s if you play **C** is *larger* than the cost of cooperation

For constant synergy factor ***r*** (Liner PGG), the above conditions imply $\boxed{r > n}$

Social dilemma exists only for *1<r<n*

C's will always be favored (cooperation is incentivized) if **r>n**

Non-linear PGG

Synergy factor is a function of number of contributors ***k***: $r \to r(k)$

$$\pi_C(k+1) - \pi_C(k) > c ==> \boxed{(k+1)r(k+1) - kr(k) > n}$$

Assuming r(k) is a linear function of ***k***: $\boxed{r(k) = r_1 + \frac{k-1}{n-1}(r_n - r_1)}$

$$\left.\begin{array}{l} r(1) = r_1 \\ r(n) = r_n \end{array}\right\}$$

$r_n > r_1 \Rightarrow$ Economies of scale ⬆Contributions ➔ r(k)⬆

$r_1 > r_n \Rightarrow$ Diminishing returns ⬆Contributions ➔ r(k)⬇

$$(k+1)r(k+1) - kr(k) > n \longrightarrow \boxed{k > k_* = \frac{(n-1)(n-r_1)}{2(r_n - r_1)}}$$

Cooperation is the rational choice when $k > k_*$

Defection is the rational choice when $k < k_*$

Non-linear PGG

1. ***Both*** full cooperation ***and*** full defection are rational if

$$\pi_C(n) - \pi_C(n-1) > c \Rightarrow nr_n - (n-1)r(n-1) > n \Rightarrow \boxed{2r_n > n + r_1}$$

$$\pi_D(0) > \pi_C(1) \Rightarrow r_1 < n \Rightarrow \boxed{2r_1 < n + r_1}$$

Combining these two inequalities ➔ $\boxed{2r_n > n + r_1 > 2r_1}$

➔ ***Both*** full **C** & full **D** can be rational only for economies of scale: $\boxed{r_n > r_1}$

2. ***Neither*** full cooperation ***nor*** full defection are rational if

$$2r_n < n + r_1 \qquad \textit{and} \qquad 2r_1 > n + r_1$$

Combining these two inequalities ➔ $\boxed{2r_1 > n + r_1 > 2r_n}$

➔ ***Neither*** full **C** nor full **D** are rational only for diminishing returns: $\boxed{r_n < r_1}$